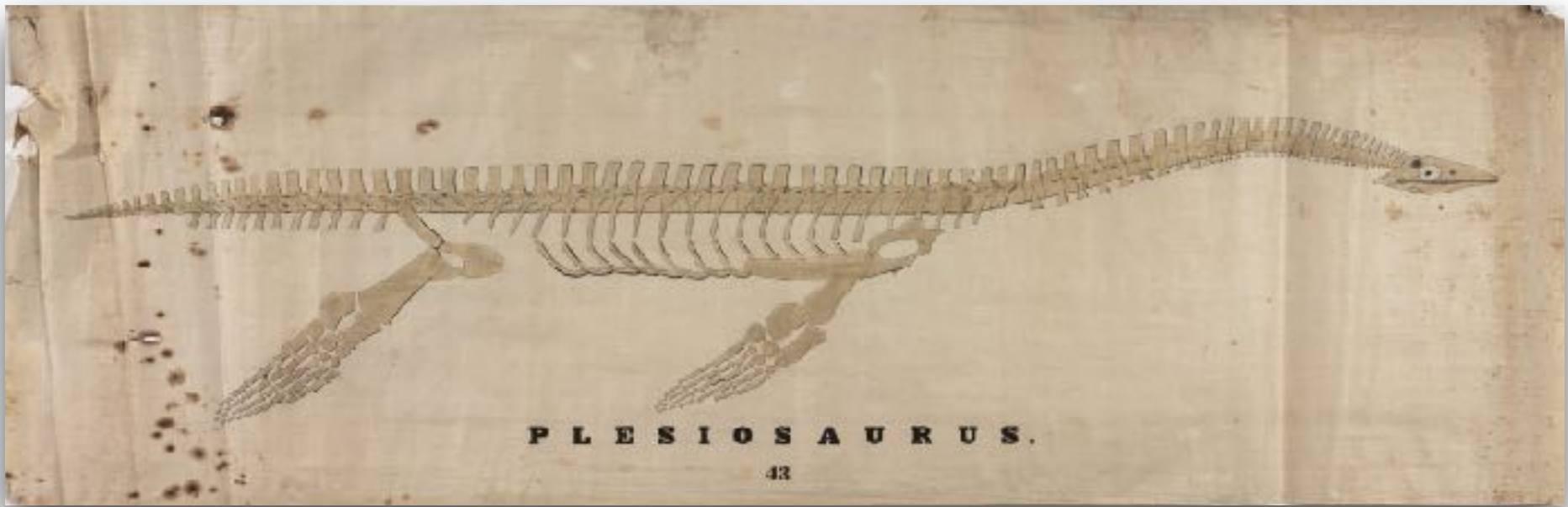
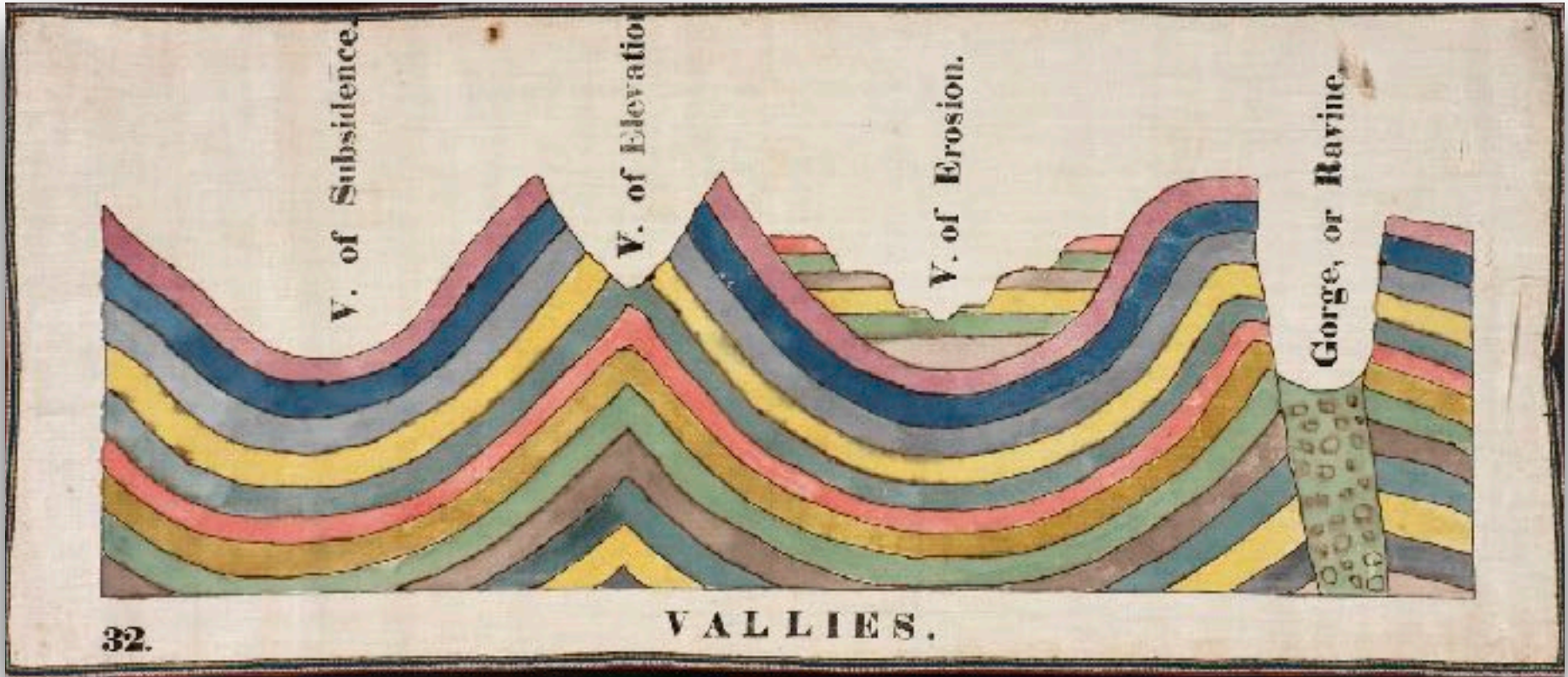
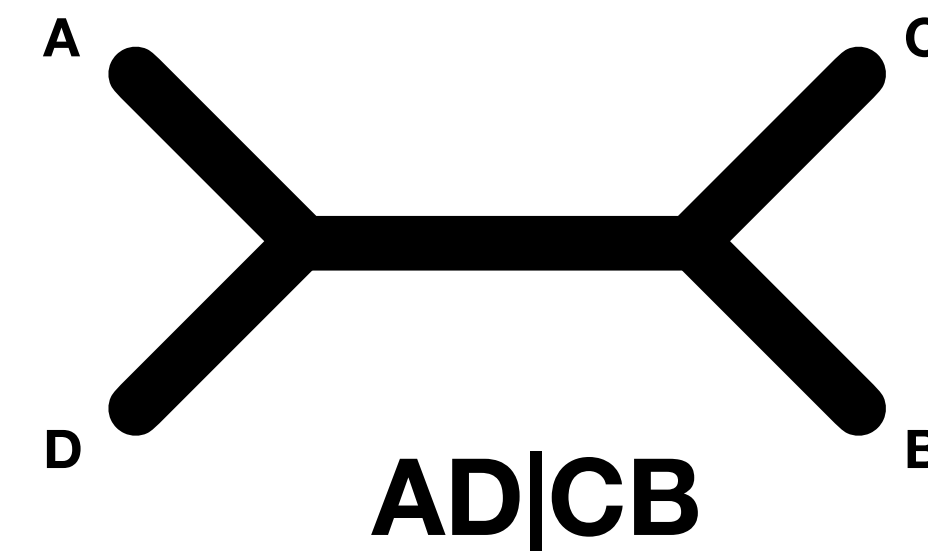
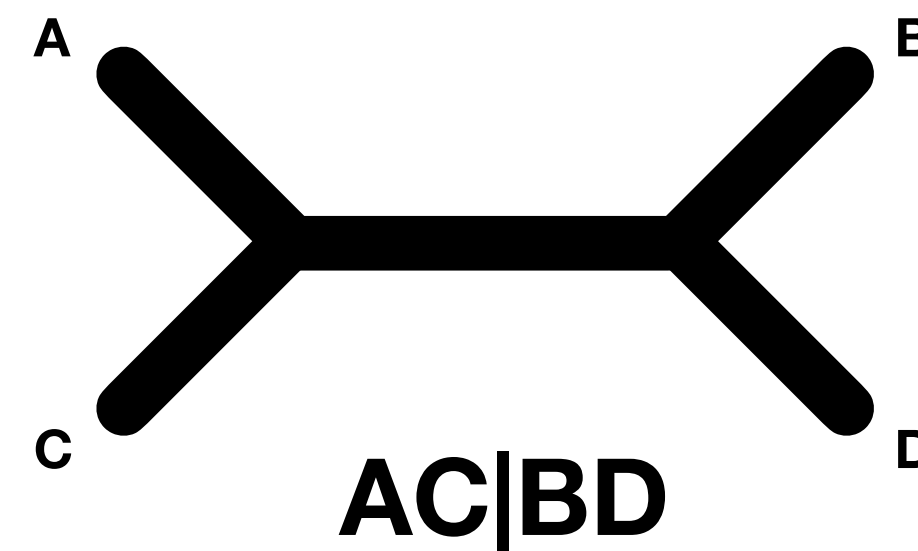
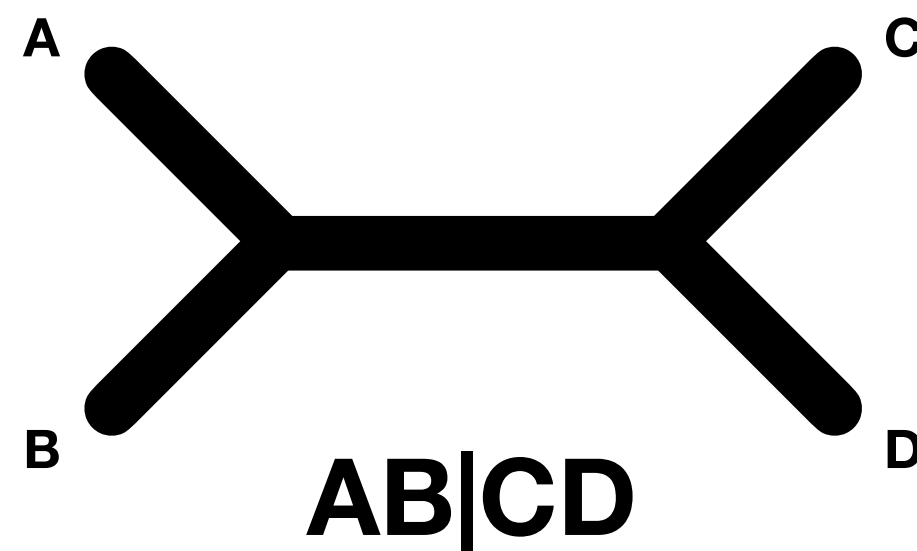
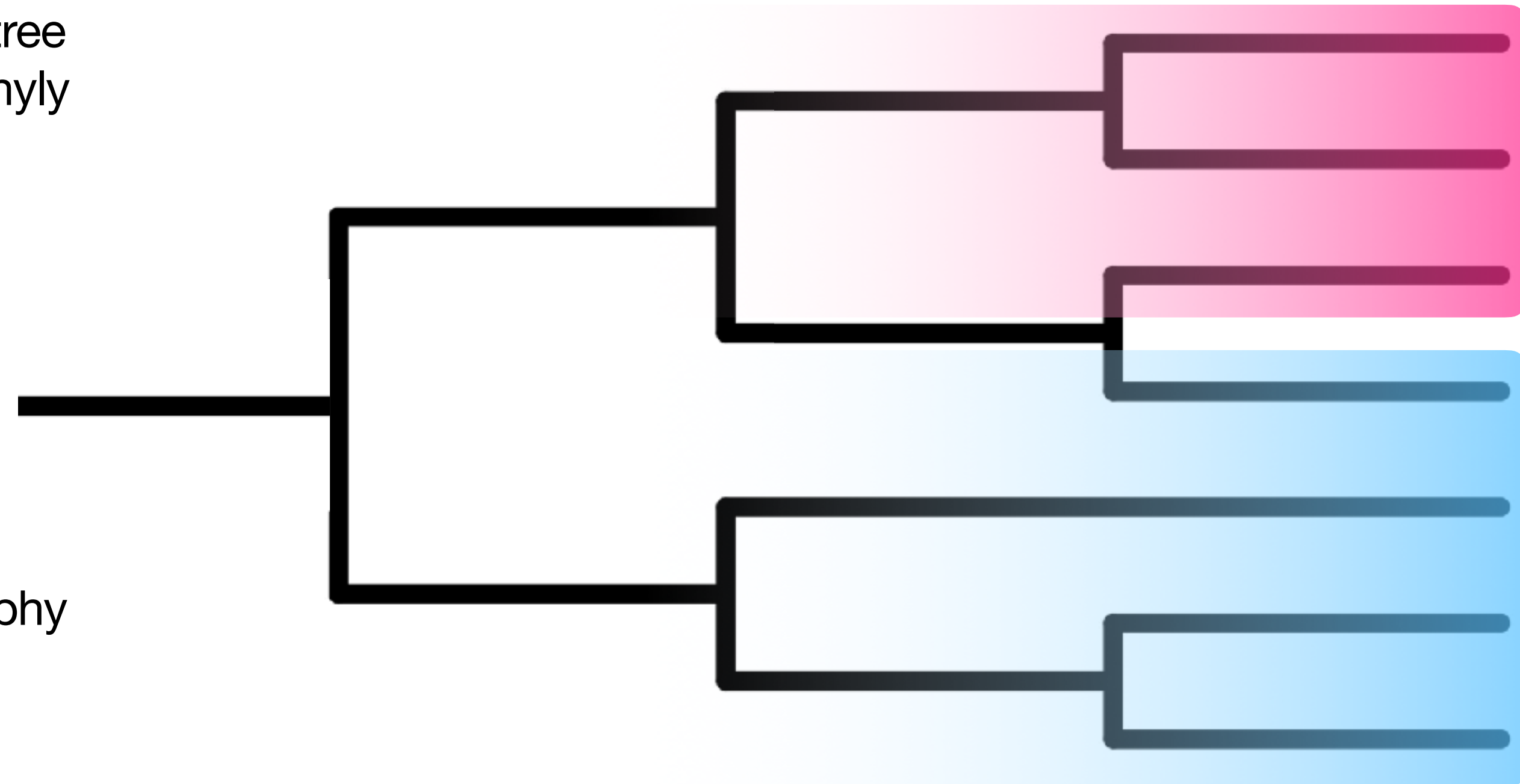




halftime recap ...

A QUICK RECAP:

root / rooted *versus* unrooted tree
 monophily / polyphyly / paraphyly
 internal / terminal node
 bipartitions
 topology / branchlengths
 cladogram
 phylogram
 chronogram (or timetree)
 dichotomy / polytomy
 clade
 synapomorphy - sinplesiomorphy
 ingroup and outgroup
 mutations and substitutions
 quartet
 homoplasy / homology
 molecoular clock hypothesis



Orthology inferences are inferred **pairwise**. When considering **multiple species**, we should move to the concept of ... **orthogroup**.

An orthogroup is **a set of genes from multiple species** that all **descend from a single gene in the last common ancestor** of those species.

An orthogroup is defined by a **reference speciation event**.

It includes all **orthologs** & **in-paralogs** that arose after the speciation events connecting the species in the comparison.

A single orthogroup can contain:

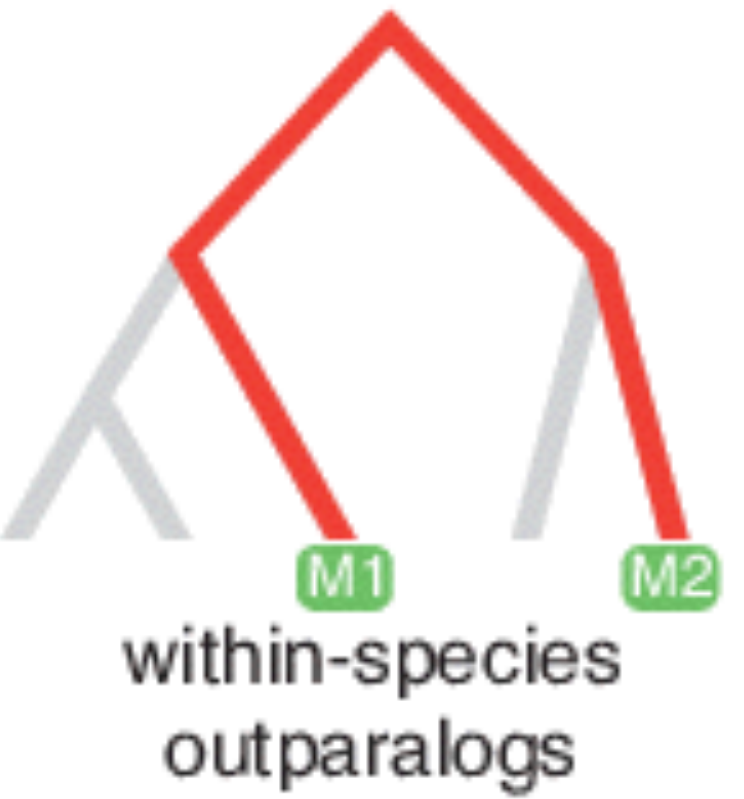
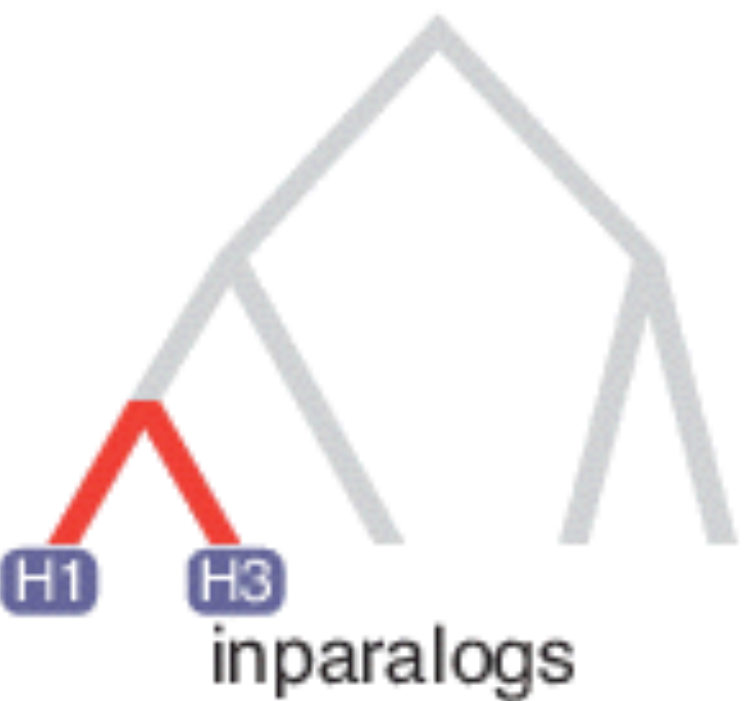
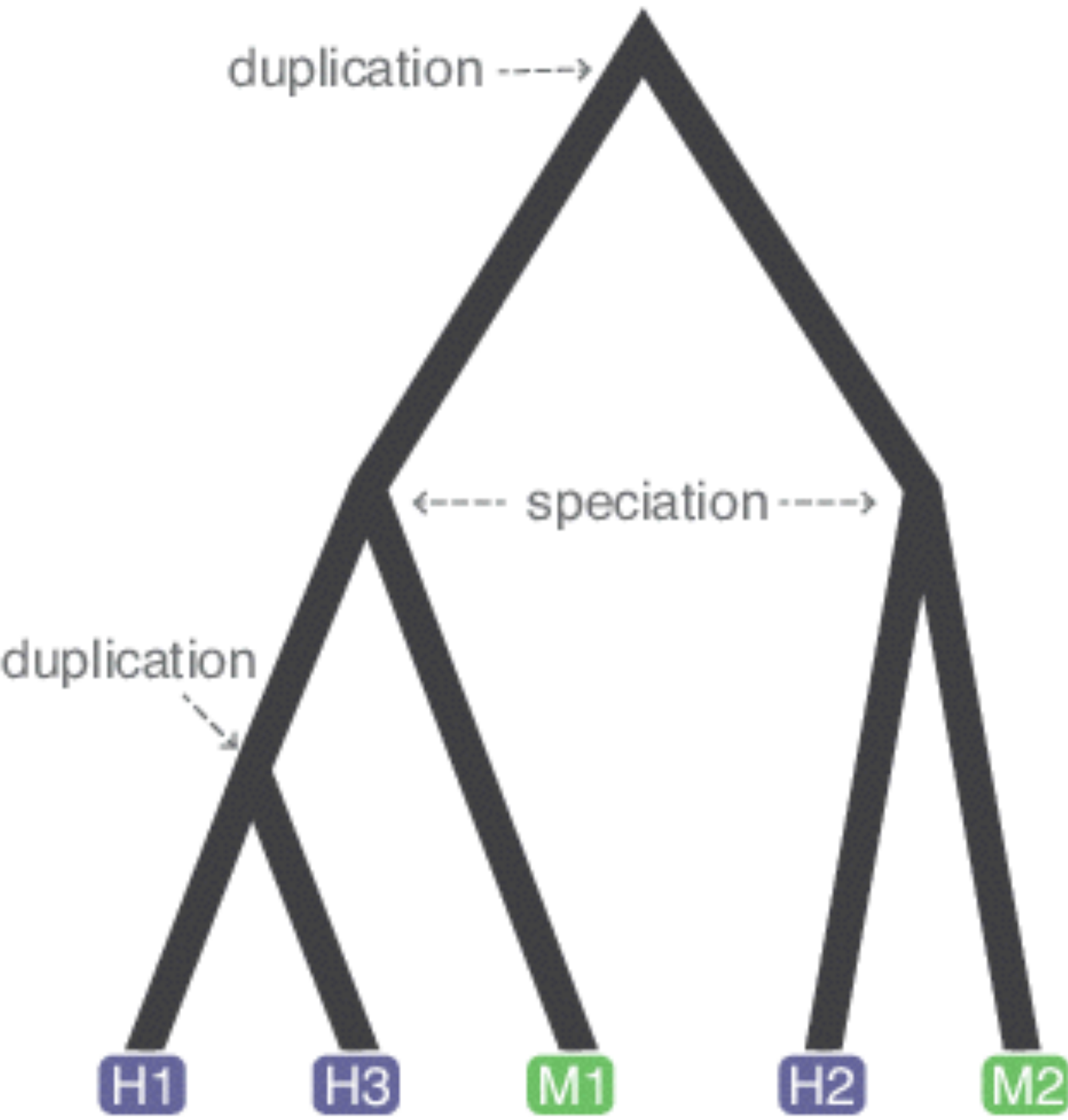
- **one-to-one relationships**
- **one-to-many relationships** (where a gene duplicated in one lineage)
- **many-to-many rel.** (where duplications occurred in multiple lineages).

Thus an orthogroup is an **homology** group.

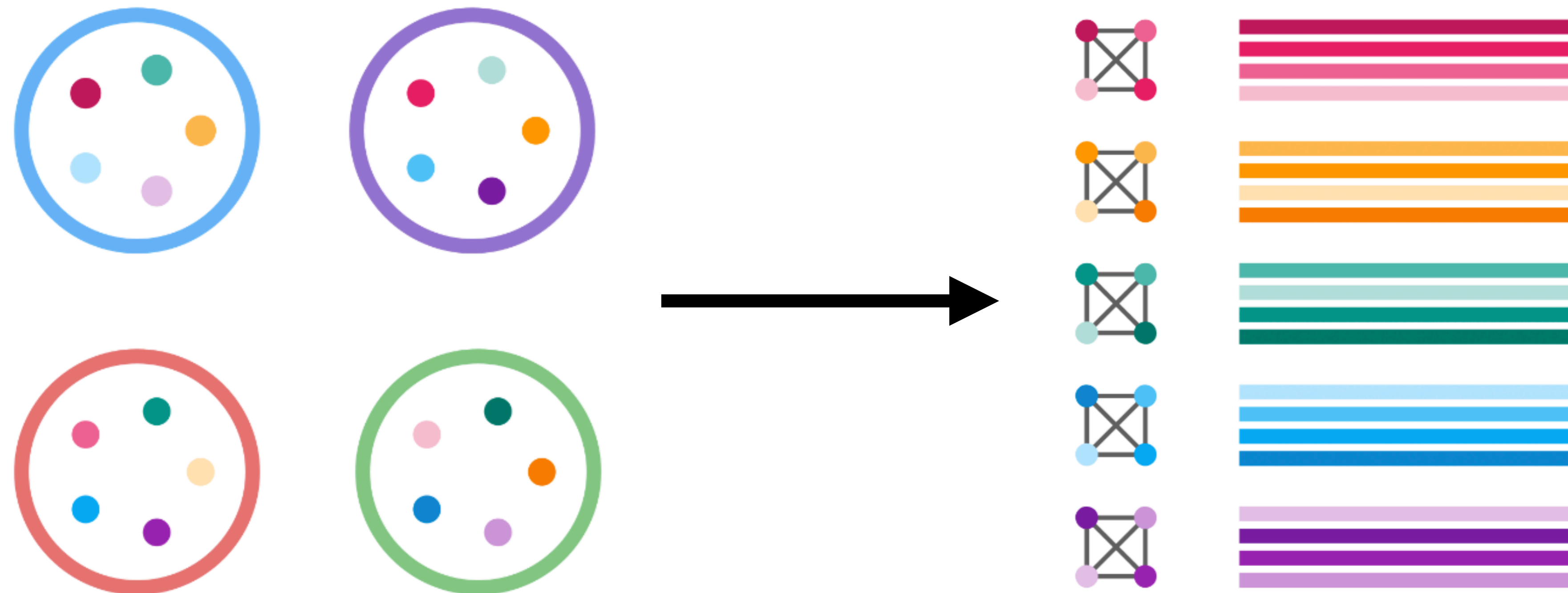
BEWARE:

Orthogroups **do not separate orthologs** and **in-paralogs** but group all genes descended from a single one in the LCA of the studied species.

... **out-paralogs** - gene duplications that occurred before the LCA - are **not included in an orthogroup**, as they belong to a different orthogroup.



OK, I have my orthogroups ... 😎



and now what?!

orthologous genes... homologous sites!

species 1	AGGATCTGCAATTGCTCTTCTAATCTGTCTGATCAGGAT
species 2	AGG-----AATTGCTCTTCTAATCTGTCT---CAGGAT
species 3	AGGATCTGCAATTGC---TCTAATCTGTCTGATCAGGAT
species 4	AGAAATCTGCAATTGCTCTTCTGATCTGTCTGATCACGAT
species 5	AGGATCTGC---TGCTCTTCTGATCTGTCTGATCAGGAT

The goal of alignment is to identify which **positions** are **homologous**.

That is, their **evolutionary history** reflect the **species relationships**!

Distance-based methods

Distance-based phylogenetic trees are based on the **total number of evolutionary changes** between pairs of sequences.

Starting from the alignment, these methods look at all possible pairs of the aligned sequences and count how many characters are different at each position. These pairwise differences are represented in a **distance matrix**.

These methods are now used for **exploratory analysis** of large datasets before more intensive tree building using character-based methods.

Character-based methods

Character-based methods compare all sequences by **considering one character** (nucleotides or amino acids) in the alignment **at a time**.

Because character-based methods incorporate **evolutionary models** they are more accurate than distance-based methods.

Character-based phylogenetic inference is really about tree-scoring, not tree-finding .. 

Maximum Likelihood $P(D|M)$

probability of observed data given a model

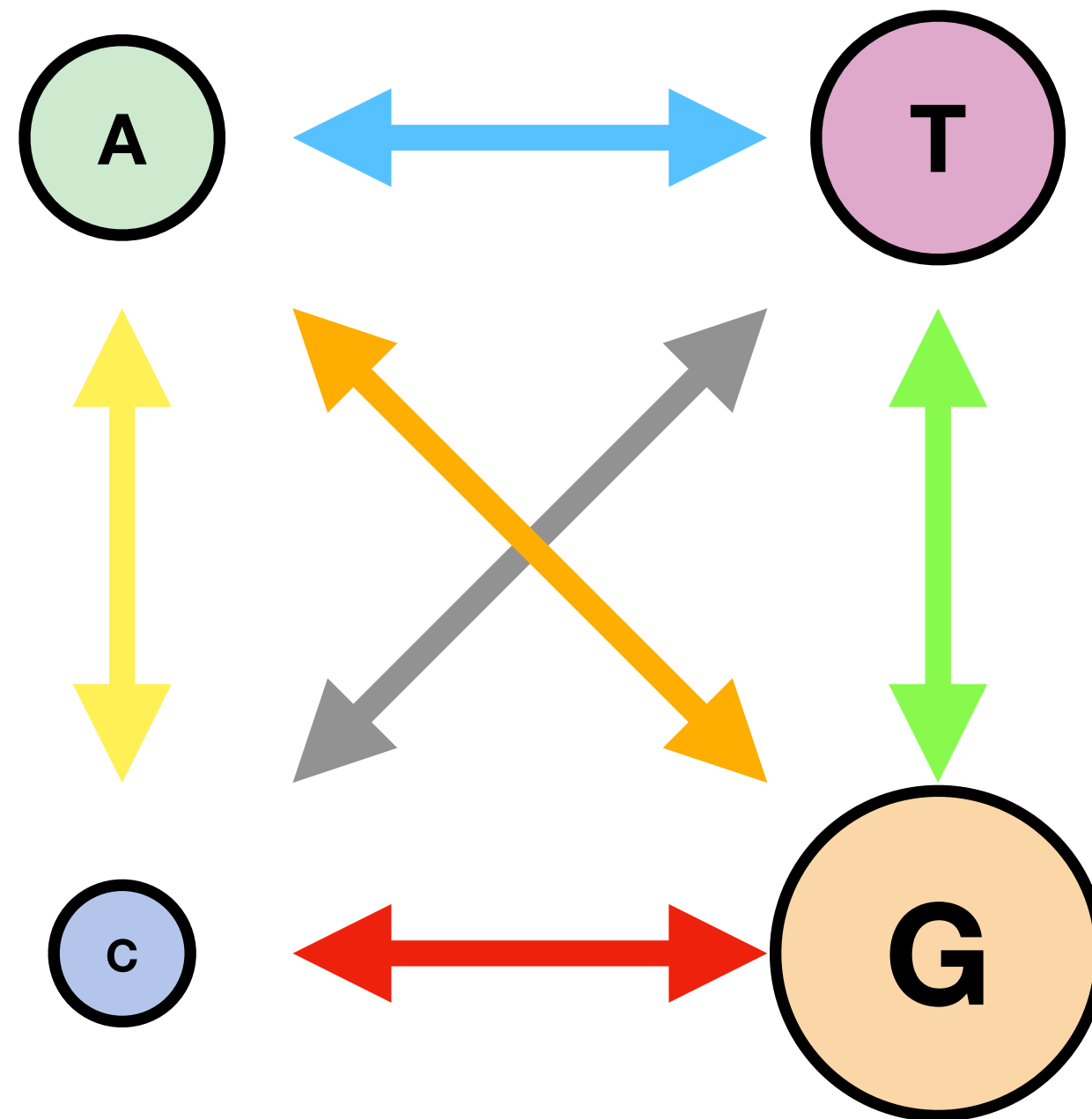
Bayesian Inference $P(M|D)$

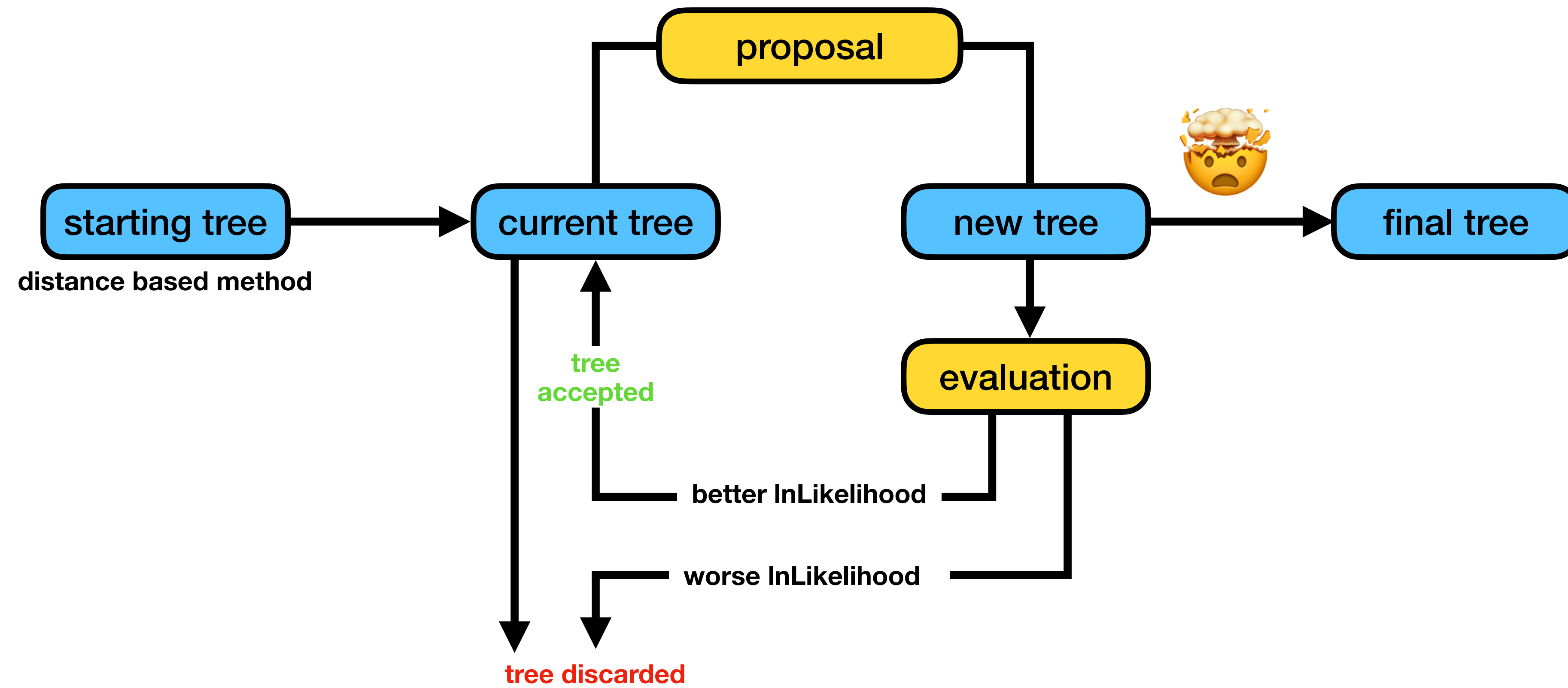
probability the model is correct given the data

GTR (General Time Reversible):

It includes a full set of six substitution rate parameters (one for each pair of nucleotides) and allows any set of equilibrium base frequencies.

All simpler models (JC, HKY, TN93, etc.) are special cases of GTR with certain rates constrained equal.





proposal

Nearest Neighbour Interchange (NNI) is a local search method that makes small, localized changes to the tree. It is the simplest and fastest tree rearrangement algorithm!

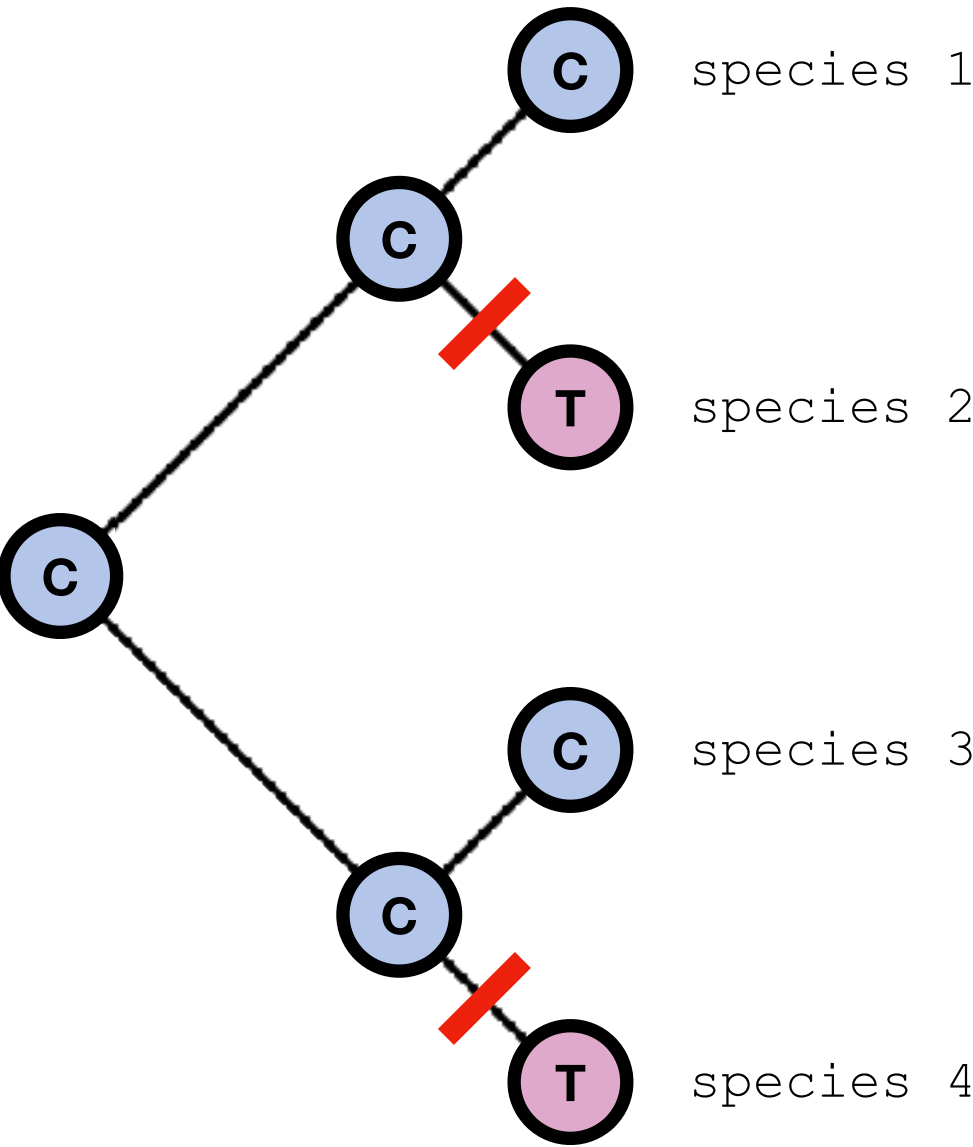
Subtree Pruning and Regrafting (SPR) is a more extensive tree rearrangement method than NNI. It moves entire subtrees rather than just swapping immediate neighbors.

Tree Bisection and Reconnection (TBR) is the most aggressive of these tree rearrangement methods and explores a much larger space of possible tree topologies. It has the lesser chance of local optima!

evaluation

		1	2	3	4	5	6	7	8
species 1		A	G	G	A	C	C	G	A
species 2		A	C	T	T	T	C	G	G
species 3		A	G	G	A	C	C	T	T
species 4		A	C	G	T	T	C	T	T

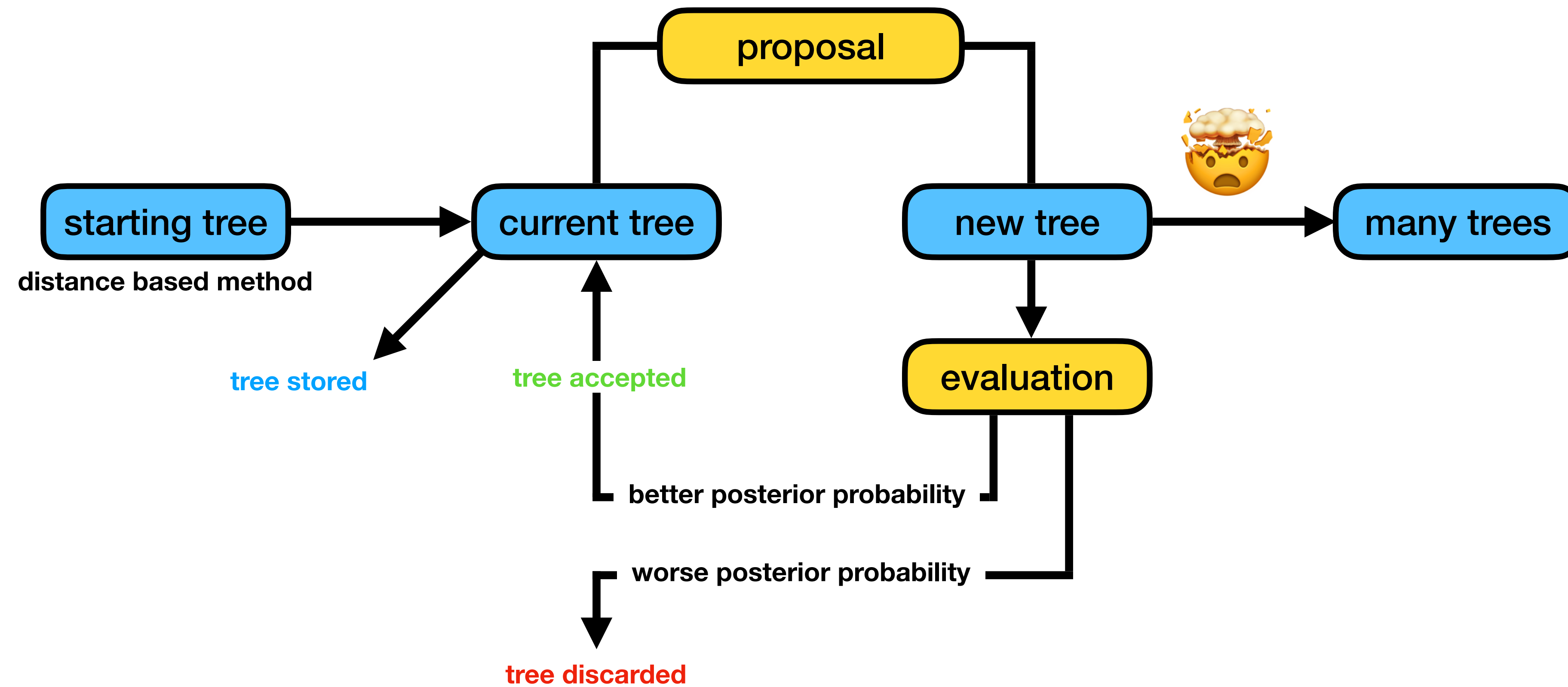
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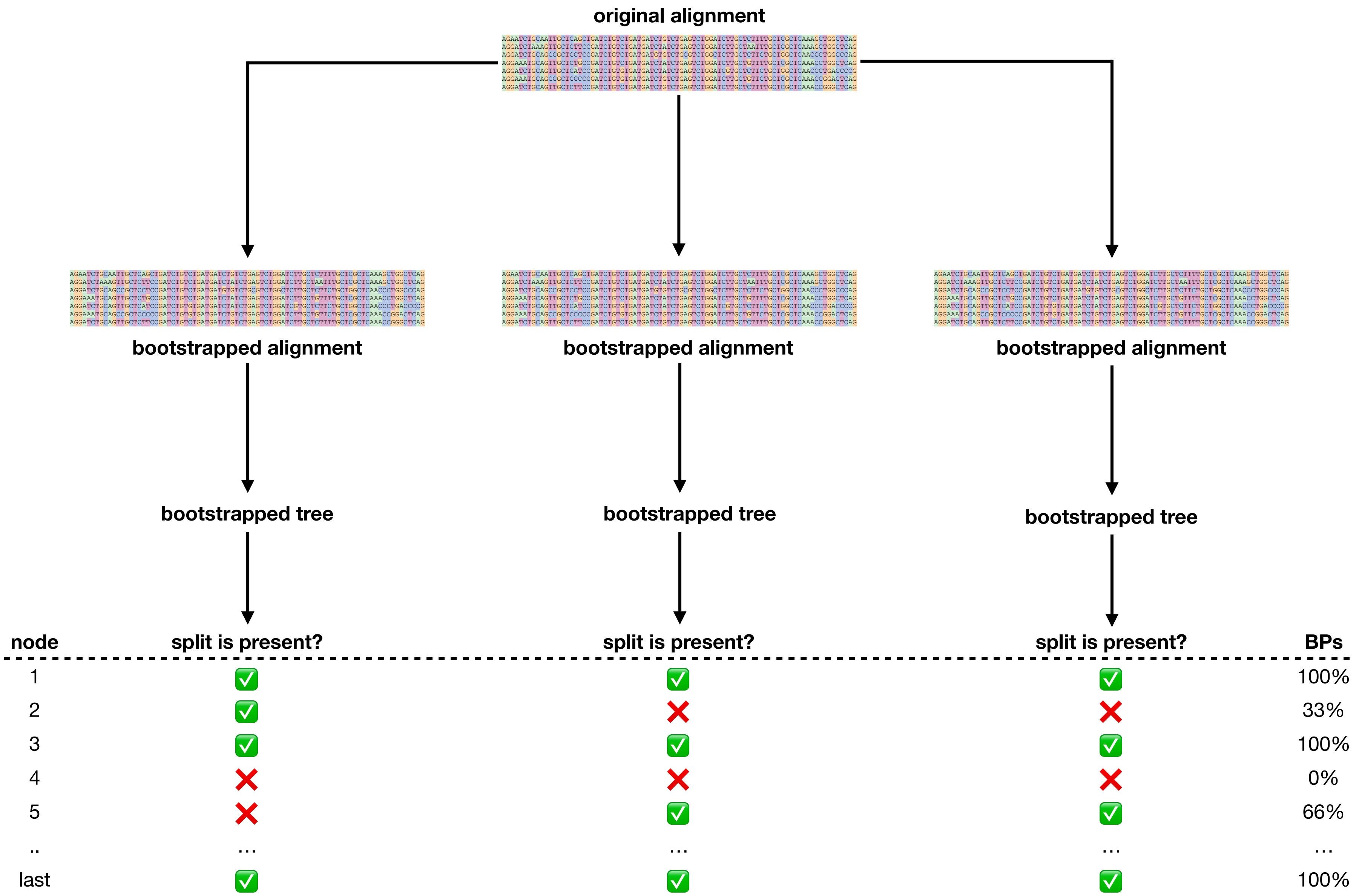


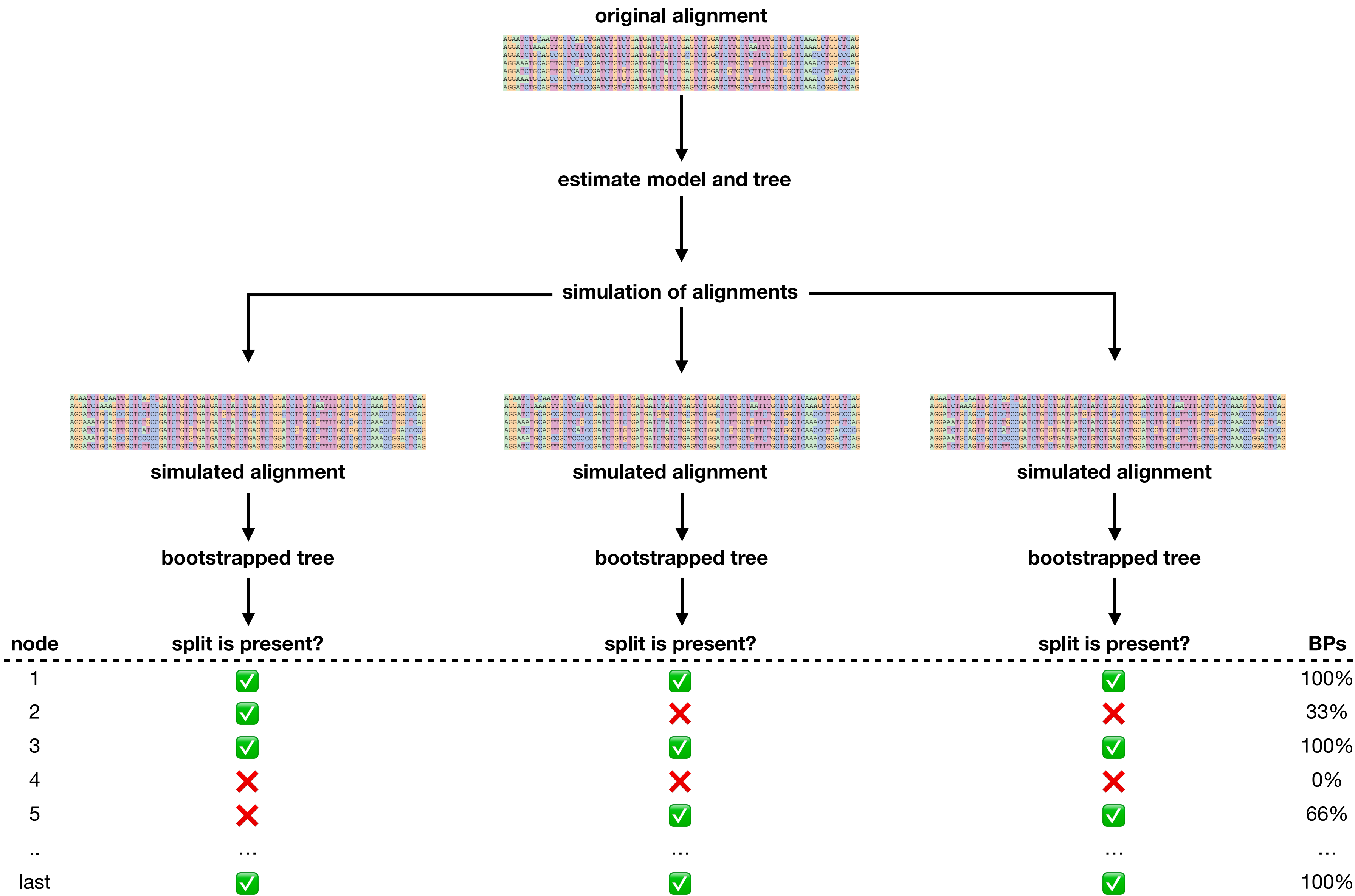
$$Q = \begin{pmatrix} & \textbf{A} & & & \\ -(\alpha\pi_G + \beta\pi_C + \gamma\pi_T) & & \alpha\pi_G & & \\ \alpha\pi_A & & -(\alpha\pi_A + \delta\pi_C + \epsilon\pi_T) & & \\ \beta\pi_A & & \delta\pi_G & & \beta\pi_C \\ \gamma\pi_A & & \epsilon\pi_G & & \delta\pi_C \\ & & & -(\beta\pi_A + \delta\pi_G + \eta\pi_T) & \\ & & & \eta\pi_C & \\ & & & & \gamma\pi_T \\ & & & & \epsilon\pi_T \\ & & & & \eta\pi_T \\ & & & & -(\gamma\pi_A + \epsilon\pi_G + \eta\pi_C) \end{pmatrix} \begin{matrix} \textbf{A} \\ \textbf{G} \\ \textbf{C} \\ \textbf{T} \end{matrix}$$

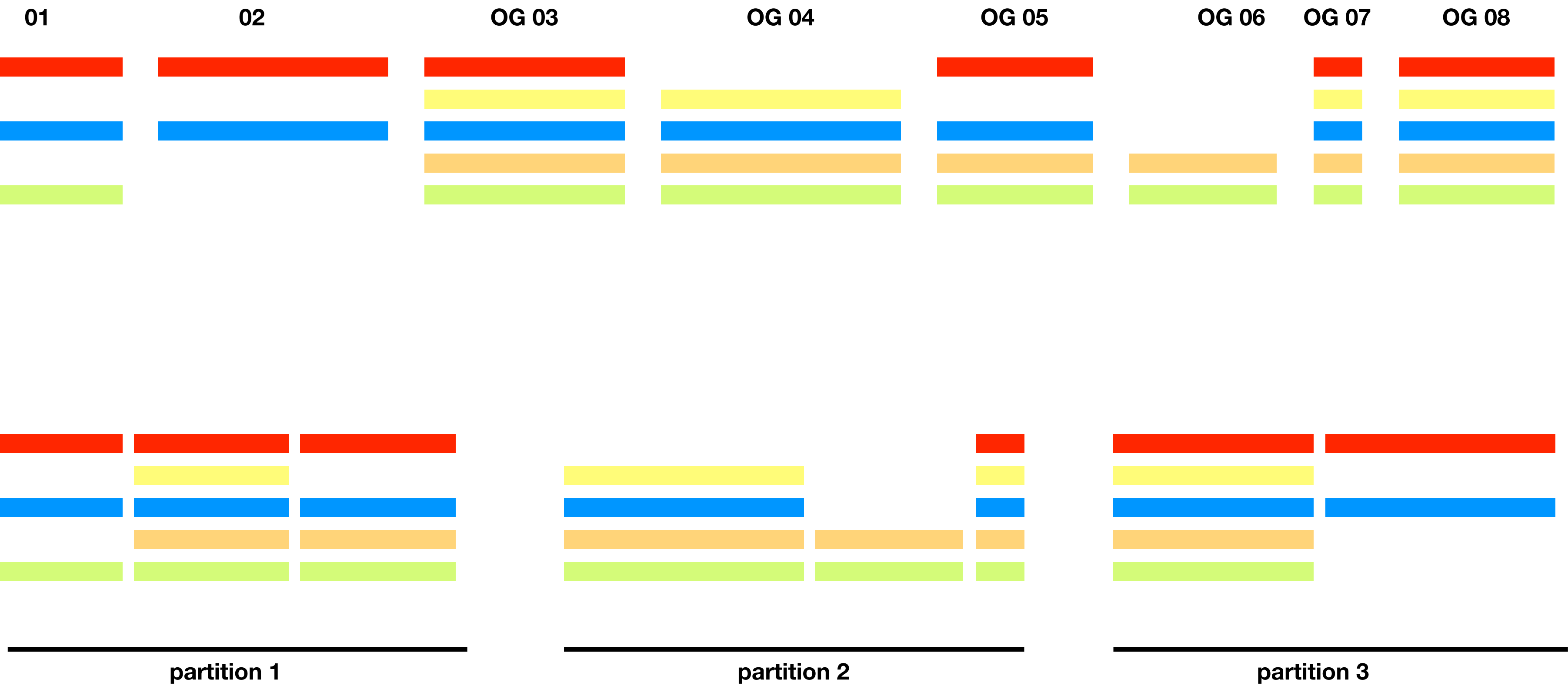
$$P_{\text{CT}} = r_{\text{CT}}\pi_{\text{T}}$$

$$P_{\text{site4}} = P_{\text{CT}} \times P_{\text{CT}} \times P_{\text{C}} \times P_{\text{C}} \times P_{\text{C}} \times P_{\text{C}}$$









Why do we merge gene into partitions?

Avoiding over-parameterization: assigning a unique model to every possible partition can lead to overfitting, where the model captures noise rather than true evolutionary signals.

Enhancing statistical power: combining partitions with similar evolutionary patterns increases the amount of data informing each parameter estimate, resulting in more robust and reliable inferences.

Improving computational efficiency: fewer partitions reduce the number of parameters to estimate, leading to faster computations and more efficient analyses.

How are partitions merged?

Because adding a parameter (dimension) to a model will always ensure a maximum likelihood at least as large as without the parameter, some penalty must be imposed when parameters are added.

How large this penalty should be is not easy to define, which has led to many different possible criteria, e.g., the **AIC** (Akaike information criterion; Akaike 1974), **AICc** (second-order AIC; Sugiura 1978; Hurvich and Tsai 1989), and **BIC** (Bayesian information criterion; Schwarz 1978).

Inadequate partitioning can lead to biased phylogenetic estimates if too few partitions are used (**under-partitioning**) or can overfit the data if too many partitions are used (**over-partitioning**).

FINISH